MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NUMERICAL INDICATOR

TYPE 8423

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

Ⓐ DESCRIPTION: Gas filled, cold cathode, glow discharge (neon)
See figure 1
Mounting position: Any

Ⓐ ABSOLUTE RATINGS:

| Parameter: | Ebb | Ik(individual) | Rp | TA | Barometric pressure, reduced |
|---|---|---|---|---|---|
| Unit: | V dc | mA | Ohms | °C | mmHg |
| Maximum: | --- | 4.5 | --- | +55 | 35 |
| Minimum: | 170 | --- | --- | -20 | --- |
| TEST CONDITIONS: | 170 | --- | 6.8 kΩ ±1% | --- | --- |

GENERAL:

Ⓐ Qualification - Not required



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| Method | Requirement or test | Notes | Conditions | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | | |
| 3347 | Ionization voltage (1) (with illumination) | | Ambient illumination = 5 to 50 ftC | Ebb | --- | 170 | V dc |
| 3333 | Uniformity of cathode glow | 1 | Amibent illumination = 5 to 50 ftC | --- | --- | --- | --- |
| 1256 | Electrode current (cathode) | 1 | Individual cathodes | Ik | 1.5 | 4.0 | mA |
| | Quality conformance inspection, part 2 | | | | | | |
| 1002 | Barometric pressure, reduced | 2,3 | eb = 170 V; pressure = 35 mmHg | --- | --- | --- | --- |
| 1031 | Variable-frequency vibration | 4,5 | | --- | --- | --- | --- |
| 1041 | Shock | 4,6 | 250 G; duration = 1.0 ±0.5 ms | --- | --- | --- | --- |
| --- | Light output | 7 | Light individual cathodes | --- | 1 | --- | ftC |
| --- | Alinement, digits | 8 | | --- | --- | ±3 | Degrees |
| Ⓐ1105 | Permanence of marking | | | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | |
| --- | Life test (1) | 9 | Group C; sequential counting; 1 Hz; t = 1,000 hours (min) | --- | --- | --- | --- |
| --- | Life-test (1) end points: | | | | | | |
| 3333 | Uniformity of cathode glow | | | --- | --- | --- | --- |
| 1256 | Electrode current (cathode) | | | Ik | 1.3 | 4.3 | mA |
| --- | Light output | | | --- | 0.8 | --- | ftC |
| --- | Life test (2) | 10,11 12 | Group C; one numeral cathode glowing steadily; 1,000 hours (min) | --- | --- | --- | --- |
| --- | Life test (2) end points: | | | | | | |
| 3333 | Uniformity of cathode glow; numeral under test | 1 | | --- | --- | --- | --- |
| 1256 | Electrode current (cathode); numeral under test | | | Ik | 1.3 | 4.5 | mA |
| --- | Light output; numeral under test | | | --- | 0.8 | --- | ftC |



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NOTES:

1. No glow shall be permitted on any tube part other than the numeral under test.

2. The tubes shall be tested by using a 60 Hz sine wave, full wave rectified and unfiltered, having the indicated peak value.

3. Pins 1 and 8 shall not be used for this test.

4. No permanent shorts or opens shall be permitted.

5. On each plane the 10 cathodes shall be sequentially energized at the test conditions.

6. Shock test shall be made on any machine capable of producing a half-sine wave-shock waveform of the specified duration and amplitude. No voltages shall be applied during this test. Each tube shall be subjected to a total of 20 shocks, i.e.; five shocks in each of positions X, Y, Z plus, and Z minus in any sequence. This is a destructive test. Tubes used in the destructive tests are not to be shipped as part of a contract.

7. Light output of each numeral is measured with a foot-candle (ftC) meter and an eye-corrected photronic cell. The entire cell area is illuminated and is coupled to the tube by a light-tight cylinder so that the face of the cell is 1.5 inches (38.10 mm) from the tube base.

8. Centerline of numerals, drawn through the top of the numeral support posts, shall be within the tolerance specified hereon in relation to the line drawn on the base of the tube through pins 1 and 8.

9. Tubes shall be energized with a duty cycle greater than 90 percent.

Ⓐ 10. Group C, life-test sampling plan, and Group C, establishment of prerelease eligibility (MIL-E-1 paragraphs) should be used with the following exceptions:

a. Group C life-test sampling plan.

(1) The first sample size = 3.
(2) The first cumulative sample size = 3.
(3) The second sample size = 3.
(4) The second cumulative sample size = 6.

b. Group C, establishment of prerelease eligibility.

(1) A minimum of 15 tubes shall-------etc.
(2) Replace all reference to the current 4-tube sample with current 3-tube sample.
Replace all reference to the last 16 with the last 15.

11. One tube shall have the top numeral cathode glowing steadily. One tube shall have the bottom numeral cathode glowing steadily. One tube shall have the "0" numeral cathode glowing steadily.

12. The life-test sample from the first lot each month shall continue on life test for an additional 1,000 hours (2,000 hours total life-test time). 2,000-hour end points shall be the same as the 1,000-hour end points, life test (2). Failure of this 2,000-hour life test shall result in the following action. Acceptance of subsequent lots shall be dependent on each lot passing the 2,000-hour life test until such time as three consecutive lots pass the 2,000-hour life test. At this time acceptance will revert to the 1,000-hour life test with only the first lot of each month continuing to 2,000 hours as before.



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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 99
DLA - ES

User activities:
Navy - AS, OS, MC, CG, SH
Air Force - 11

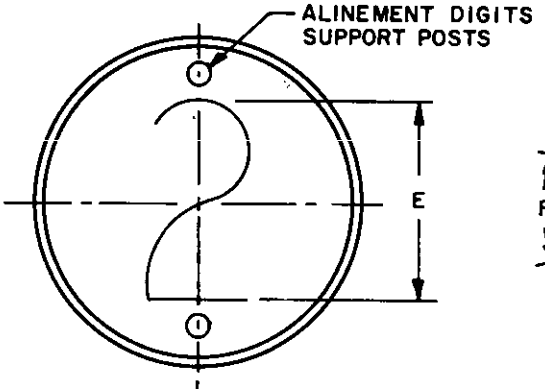
Agent:
DLA - ES

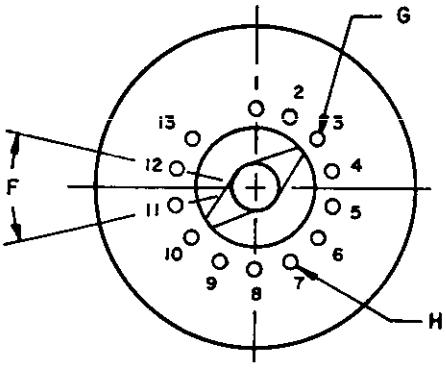
Preparing activity:
Army - ER

(Project 5960-3343)

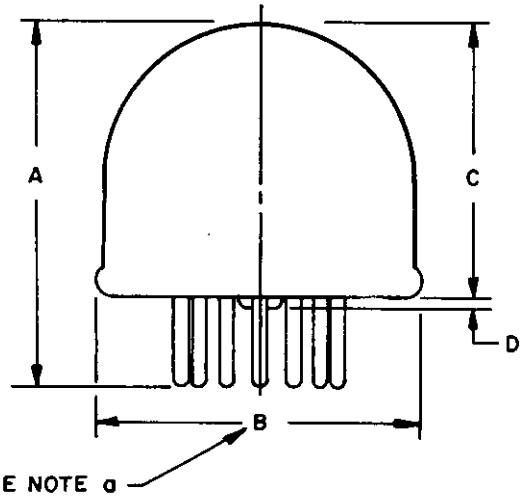


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BOTTOM VIEW




SEE NOTE a

1234567890

NUMERAL DESIGN

| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 2 | | |
| A | --- | 1.522 (38.66) |
| B | --- | 1.350 (34.29) |
| C | --- | 1.212 (30.78) |
| D | --- | .064 (1.63) |
| E | .790 (20.07) | .820 (20.83) |
| Reference dimensions (see note b) | | |
| F | 25.7° | |
| G | .328 (8.33) pin radius, applies to all pins | |
| H | .040 (1.02) pin diameter | |

| PIN NO. | ELEMENT |
|---|---|
| 1 | INT CON |
| 2 | ANODE |
| 3 | k0 |
| 4 | k9 |
| 5 | k8 |
| 6 | k7 |
| 7 | k6 |
| 8 | INT CON |
| 9 | k5 |
| 10 | k4 |
| 11 | k3 |
| 12 | k2 |
| 13 | k1 |

NOTES:
a. Applies to entire length of bulb.
b. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 8423.



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